Precept: Bayesian Exercises

November 3, 2026

Problems

As a reminder, here is Bayes' rule

$$P(x \mid y) = \frac{P(y \mid x)P(x)}{P(y)}$$

and to compute $P(y)$, we can use marginalization and conditionalization to write

$$P(y) = \sum_x P(x, y) = \sum_x P(y \mid x)P(x)$$

1. **The trick coin.** A bag holds two coins. One is fair, and the other has heads on both sides. You pull a coin out at random and flip it.
   (a) It lands heads. What is the probability that you pulled out the two-headed coin?
   (b) You flip the same coin again and it lands heads again. What is the probability now?

2. **The positive test.** A rare condition affects 1% of the population. A test for it comes back positive 99% of the time when someone has the condition, and gives a false positive 5% of the time when someone does not.
   (a) You take the test and it comes back positive. What is the probability that you actually have the condition?

3. **The two-child puzzle.** A family has two children. Assume each child is equally likely to be a girl or a boy, independently of the other.
   (a) You learn that *at least one* of the children is a girl. What is the probability that both are girls?
   (b) Instead, you learn that the *older* child is a girl. What is the probability that both are girls?

4. **Rewriting variance.** We saw that the formula for variance is $\mathbb{E}\big[(x - \mathbb{E}[x])^2\big]$. This is a bit cumbersome. To make our lives easier, show that
   $$\mathbb{E}\big[(x - \mathbb{E}[x])^2\big] = \mathbb{E}\big[x^2\big] - \mathbb{E}\big[x\big]^2$$
   (*Hint:* Think back to the properties of linear functions. Is the expectation linear? (*Another hint:* It is!))

Solution

Question 1

Let U be "two-headed coin" (it's U for "Unfair") and F be "fair coin," so $P(U) = P(F) = \frac{1}{2}$, with $P(H \mid U) = 1$ and $P(H \mid F) = \frac{1}{2}$.

Part (a)

This is just Bayes' rule!

$$P(U \mid H) = \frac{P(H \mid U)P(U)}{P(H)}$$

To compute $P(H)$, we can use marginalization and the product rule to get

$$P(H) = P(H \mid U)P(U) + P(H \mid F)P(F) = (1)\left(\tfrac{1}{2}\right) + \left(\tfrac{1}{2}\right)\left(\tfrac{1}{2}\right) = \tfrac{3}{4}$$

so that

$$P(U \mid H) = \frac{P(H \mid U)P(U)}{P(H)} = \frac{1/2}{3/4} = \frac{2}{3}$$

Part (b)

Given the coin, the flips are independent, so $P(HH \mid U) = 1$ and $P(HH \mid F) = \frac{1}{4}$. Then

$$P(U \mid HH) = \frac{(1)\left(\frac{1}{2}\right)}{(1)\left(\frac{1}{2}\right) + \left(\frac{1}{4}\right)\left(\frac{1}{2}\right)} = \frac{1/2}{5/8} = \frac{4}{5}$$

Each head makes us more confident in the two-headed coin. Notice that a single tails would immediately drop $P(U)$ to 0.

Question 2

Let C mean "has the condition" and $\bar{C}$ mean "does not." We are given

$$P(C) = 0.01, \qquad P(+ \mid C) = 0.99, \qquad P(+ \mid \bar{C}) = 0.05$$

Part (a)

We can just use Bayes' rule!

$$P(D \mid +) = \frac{P(+ \mid D)\, P(D)}{P(+)}$$

By substituting in the values given to us, we can compute $P(+)$ as

$$P(+) = P(+, D) + P(+, \bar{D}) = P(+ \mid D)P(D) + P(+ \mid \bar{D})P(\bar{D}) = (0.99)(0.01) + (0.05)(0.99) = 0.0594$$

Therefore, we can write

$$P(D \mid +) = \frac{0.0099}{0.0594} \approx 0.17$$

Funny how that's not 99%, right?

Question 3

Listing the children as (older, younger), the four outcomes GG, GB, BG, BB each have probability $\frac{1}{4}$.

Part (a)

Again! This is Bayes' rule!

$$P(GG \mid \text{at least one } G) = \frac{P(\text{at least one } G \mid GG) \cdot P(GG)}{P(\text{at least one } G)}$$

Given that there is "at least one girl", we know that there are three possible states of the world $\{GG, GB, BG\}$, so

$$P(\text{at least one } G) = \frac{3}{4}$$

Notice that $P(\text{at least one } G \mid GG) = 1$ and $P(GG) = \frac{1}{4}$. Hence, we get Since GG is contained in this event, $P(GG, \text{at least one } G) = P(GG) = \frac{1}{4}$. By the product rule,

$$P(GG \mid \text{at least one } G) = \frac{P(\text{at least one } G \mid GG) \cdot P(GG)}{P(\text{at least one } G)} = \frac{1/4}{3/4} = \frac{1}{3}$$

Part (b)

We can intuitively reason about this one. Since the "older child is a girl", there are two possible states of the world $\{GG, GB\}$. Given they are both equal probability, the probability that both children are girls is $\frac{1}{2}$.

Question 4

Following the hint, the expectation is linear! Hence, we can do the following:

$$\begin{aligned}
\mathbb{E}\big[(x - \mathbb{E}[x])^2\big] &= \mathbb{E}\big[x^2 - 2x\mathbb{E}\big[x\big] + \mathbb{E}\big[x\big]^2\big] \\
&= \mathbb{E}\big[x^2\big] - \mathbb{E}\big[2x\mathbb{E}\big[x\big]\big] + \mathbb{E}\big[\mathbb{E}\big[x\big]^2\big] \\
&= \mathbb{E}\big[x^2\big] - 2\mathbb{E}\big[x\big]\mathbb{E}\big[x\big] + \mathbb{E}\big[x\big]^2 \\
&= \mathbb{E}\big[x^2\big] - 2\mathbb{E}\big[x\big]^2 + \mathbb{E}\big[x\big]^2 \\
&= \mathbb{E}\big[x^2\big] - \mathbb{E}\big[x\big]^2
\end{aligned}$$